Balanced Bi-Independent Pairs:
An Integer-Refined Spectral Bound and the Hypercube Case

Abstract

For an r-regular bipartite graph $G = (V_1 \sqcup V_2, E)$ with $|V_1| = |V_2| = n$, let λ denote the second singular value of its bipartite adjacency matrix. The standard eigenvalue argument gives

$$\alpha_{\mathrm{bal}}(G) \le \frac{2n\lambda}{r+\lambda} = 4\mathrm{b}h(G).$$

We give an arithmetic refinement of this bound. The improvement comes from replacing the usual Cauchy–Schwarz lower bound on a sum of squares by the exact integer minimum of that sum of squares. The resulting bound is never weaker than the standard one and is strictly stronger for a small explicit regular bipartite graph.

We also record the exact value of $\alpha_{\mathrm{bal}}(Q_d)$ for the hypercube. The hypercube conjecture of Laurent, Polak and Vargas follows directly from a 2012 theorem of Barber on balanced independent sets in the cube. Equivalently, if

$$a(0) = 0, \qquad a(2s) = 2^{2s} - \binom{2s}{s}, \qquad a(2s+1) = 2a(2s),$$

then

$$\alpha_{\mathrm{bal}}(Q_d) = a(d-1) \qquad (d \ge 1).$$

1 Introduction

Let $G = (V_1 \sqcup V_2, E)$ be a bipartite graph. A pair (A, B), with $A \subseteq V_1$ and $B \subseteq V_2$, is called *bi-independent* if there is no edge between A and B. It is called *balanced* if $|A| = |B|$. We write

$$\alpha_{\mathrm{bal}}(G) = \max\{\, |A| + |B| : (A, B) \text{ is a balanced bi-independent pair}\,\}.$$

This parameter was studied by Laurent, Polak and Vargas in their work on semidefinite approximations for bicliques and bi-independent pairs [4]. They asked whether the usual eigenvalue-based bounds can be strengthened in a way that genuinely uses the balance condition, and they formulated a concrete conjecture for the hypercube Q_d.

The present note has two independent parts. First, in Sections 2–4, we prove a general integer-refined spectral upper bound for $\alpha_{\mathrm{bal}}(G)$ in regular bipartite graphs. Second, in Section 5, we explain that the hypercube conjecture follows from Barber's theorem on balanced independent sets in the cube [1]. The hypercube argument is combinatorial and uses the cube isoperimetric theorem of Bezrukov and of Körner–Wei; it is not a consequence of the spectral refinement.

2 An integer-refined spectral bound

Let $G = (V_1 \sqcup V_2, E)$ be an r-regular bipartite graph with

$$|V_1| = |V_2| = n, \qquad 1 \le r \le n.$$

Let $M \in \{0,1\}^{n\times n}$ be the bipartite adjacency matrix of G, so that the full adjacency matrix is

$$A_G = \begin{pmatrix} 0 & M \\ M^{\mathsf{T}} & 0 \end{pmatrix}.$$

Put

$$\lambda := \sup\{\|Mx\|_2 : x \perp \mathbf{1}, \|x\|_2 = 1\},$$

with the convention that this supremum is 0 if $\mathbf{1}^\perp = \{0\}$. For $n \ge 2$, this is $\sigma_2(M)$, the second singular value of M.

For each integer a with

$$0 \le a \le \rho := \min\{n - r, \lfloor n/2 \rfloor\},$$

put

$$m_a := n - a,$$

and let j_a be the least nonnegative residue of ra modulo m_a:

$$j_a \equiv ra \pmod{m_a}, \qquad 0 \le j_a < m_a.$$

Define

$$B_{\text{int}}(n, r, \lambda) := 2 \max\left\{a \in \{0, 1, \dots, \rho\} : r^2 a^3 + n\, j_a(m_a - j_a) \le \lambda^2 a\, m_a^2\right\}.$$

The set over which the maximum is taken is nonempty, since $a = 0$ is always admissible.

Theorem 1 (Integer-refined spectral bound)**.** *For every r-regular bipartite graph $G = (V_1 \sqcup V_2, E)$ with $|V_1| = |V_2| = n$,*

$$\boxed{\alpha_{\text{bal}}(G) \le B_{\text{int}}(n, r, \lambda).}$$

Equivalently, if (A, B) is balanced bi-independent and $|A| = |B| = a$, then

$$\boxed{r^2 a^3 + n\, j_a(n - a - j_a) \le \lambda^2 a (n - a)^2.}$$

We first record the elementary integer minimization used in the proof.

Lemma 1. *Let $m \ge 1$ and $s \ge 0$ be integers. Write*

$$s = qm + j, \qquad 0 \le j < m.$$

Among all m-tuples $(x_1, \dots, x_m)$ of nonnegative integers with $x_1 + \cdots + x_m = s$, the minimum of $x_1^2 + \cdots + x_m^2$ is

$$(m - j)q^2 + j(q + 1)^2 = \frac{s^2}{m} + \frac{j(m - j)}{m}.$$

Proof. If two coordinates differ by at least 2, say $u \ge v + 2$, then replacing (u, v) by $(u - 1, v + 1)$ preserves the sum and decreases the sum of squares by

$$u^2 + v^2 - (u - 1)^2 - (v + 1)^2 = 2(u - v - 1) > 0.$$

Thus, in a minimizing tuple, all entries differ by at most 1. Since the sum is $s = qm + j$, such a tuple has exactly $m - j$ entries equal to q and j entries equal to $q + 1$. Expanding gives the displayed value. $\square$

Proof of Theorem 1. Let (A, B) be a balanced bi-independent pair with

$$|A| = |B| = a.$$

If $a = 0$, there is nothing to prove, so assume $a > 0$.

Since every vertex of A has all its r neighbors in $V_2 \setminus B$, we must have $n - a \geq r$. Also, the ra edges incident with B all end in $V_1 \setminus A$, and each vertex in $V_1 \setminus A$ is incident with at most r of these edges; hence $ra \leq r(n-a)$, so $a \leq n/2$. Therefore $a \leq \rho$ and $m := n - a$ is positive.

Let $y = \mathbf{1}_B \in \{0,1\}^n$. Since A has no edges to B, the entries of My indexed by A are equal to 0. On the remaining $m = n - a$ coordinates, the entries of My are nonnegative integers whose sum is

$$\mathbf{1}^\mathsf{T} My = (M^\mathsf{T}\mathbf{1})^\mathsf{T} y = r\mathbf{1}^\mathsf{T} y = ra.$$

Write

$$ra = qm + j, \qquad 0 \leq j < m.$$

Thus $j = j_a$. Applying Lemma 1 with $s = ra$ gives

$$\|My\|_2^2 \geq (m-j)q^2 + j(q+1)^2 = \frac{r^2a^2}{m} + \frac{j(m-j)}{m}. \tag{1}$$

We now upper-bound the same quantity spectrally. Decompose

$$y = \frac{a}{n}\mathbf{1} + y_0, \qquad y_0 \perp \mathbf{1}.$$

Because $M\mathbf{1} = r\mathbf{1}$ and $M^\mathsf{T}\mathbf{1} = r\mathbf{1}$, the two vectors

$$M\left(\frac{a}{n}\mathbf{1}\right) \quad \text{and} \quad My_0$$

are orthogonal. Hence

$$\|My\|_2^2 = \left\|M\left(\frac{a}{n}\mathbf{1}\right)\right\|_2^2 + \|My_0\|_2^2.$$

The first term is

$$\left\|\frac{ar}{n}\mathbf{1}\right\|_2^2 = \frac{r^2a^2}{n}.$$

Since $y_0 \perp \mathbf{1}$, the definition of λ gives

$$\|My_0\|_2^2 \leq \lambda^2 \|y_0\|_2^2.$$

Moreover,

$$\|y_0\|_2^2 = \left\|\mathbf{1}_B - \frac{a}{n}\mathbf{1}\right\|_2^2 = a - \frac{a^2}{n} = \frac{a(n-a)}{n}.$$

Thus

$$\|My\|_2^2 \leq \frac{r^2a^2}{n} + \frac{\lambda^2 a(n-a)}{n}. \tag{2}$$

Combining (1) and (2), with $m = n - a$, gives

$$\frac{r^2a^2}{m} + \frac{j(m-j)}{m} \leq \frac{r^2a^2}{n} + \frac{\lambda^2 am}{n}.$$

Multiplying by nm yields

$$nr^2a^2 + n\,j(m-j) \leq mr^2a^2 + \lambda^2 am^2.$$

Since $n - m = a$, this is equivalent to

$$r^2 a^3 + n\, j(m - j) \le \lambda^2 a\, m^2.$$

Substituting $m = m_a = n - a$ and $j = j_a$ gives

$$r^2 a^3 + n\, j_a(m_a - j_a) \le \lambda^2 a\, m_a^2.$$

Therefore every balanced bi-independent pair of size $2a$ satisfies the condition appearing in the definition of $B_{\text{int}}(n, r, \lambda)$. Taking the maximum over all balanced bi-independent pairs proves the theorem. $\square$

3 Comparison with the standard eigenvalue bound

Corollary 1. *For every r-regular bipartite graph G as in Theorem 1,*

$$B_{\text{int}}(n, r, \lambda) \le 2 \left\lfloor \frac{n\lambda}{r + \lambda} \right\rfloor \le \frac{2n\lambda}{r + \lambda} = 4\text{b}h(G),$$

where

$$\text{b}h(G) = \frac{n\lambda}{2(r + \lambda)}.$$

Hence the integer-refined bound is never weaker than the bound $\alpha_{\text{bal}}(G) \le 4\text{b}h(G)$.

Proof. Let $a > 0$ satisfy the defining inequality for $B_{\text{int}}(n, r, \lambda)$. Since

$$n\, j_a(m_a - j_a) \ge 0,$$

we have

$$r^2 a^3 \le \lambda^2 a(n - a)^2.$$

Dividing by $a > 0$ and taking square roots gives

$$ra \le \lambda(n - a).$$

Therefore

$$a \le \frac{n\lambda}{r + \lambda}.$$

The claim follows by maximizing over all admissible integers a. $\square$

The improvement over the standard estimate is precisely the additional nonnegative integer term

$$n\, j_a(m_a - j_a).$$

This term is absent from the continuous Cauchy–Schwarz lower bound. Whenever ra is not divisible by $n - a$, it is positive and may rule out a balanced pair size that the older estimate still permits.

4 A strict example

Example 1. *Let* $V_1 = V_2 = \mathbb{Z}_8$*, and define a bipartite graph by declaring* $i \in V_1$ *adjacent to* $j \in V_2$ *exactly when*

$$j - i \in \{0, 1, 2, 3, 5\} \pmod 8.$$

This is a 5*-regular bipartite graph, so* $n = 8$ *and* $r = 5$*.*

Its bipartite adjacency matrix is circulant. If $\omega = e^{2\pi i/8}$*, then its eigenvalues are*

$$\mu_k = 1 + \omega^k + \omega^{2k} + \omega^{3k} + \omega^{5k}, \qquad k = 0, \dots, 7.$$

Because the matrix is circulant and hence normal, its singular values are $|\mu_k|$*. A direct calculation gives*

$$\left(|\mu_0|^2, \dots, |\mu_7|^2\right) = (25, 3, 1, 3, 1, 3, 1, 3).$$

Therefore

$$\lambda = \sigma_2(M) = \sqrt{3}.$$

The standard eigenvalue bound gives

$$4\mathrm{b}h(G) = \frac{2n\lambda}{r + \lambda} = \frac{16\sqrt{3}}{5 + \sqrt{3}} \approx 4.116.$$

Since $\alpha_{\mathrm{bal}}(G)$ *is even, this only implies*

$$\alpha_{\mathrm{bal}}(G) \leq 4.$$

Now apply Theorem 1. The possible values of a *in* $B_{\mathrm{int}}(8, 5, \sqrt{3})$ *are* $0, 1, 2, 3$*. For* $a = 2$*, we have*

$$m_a = n - a = 6, \qquad j_a \equiv ra = 10 \equiv 4 \pmod 6.$$

The defining inequality would require

$$r^2 a^3 + n\, j_a(m_a - j_a) \leq \lambda^2 a\, m_a^2.$$

But here

$$r^2 a^3 + n\, j_a(m_a - j_a) = 25 \cdot 8 + 8 \cdot 4 \cdot 2 = 264,$$

whereas

$$\lambda^2 a\, m_a^2 = 3 \cdot 2 \cdot 6^2 = 216.$$

Thus $a = 2$ *is impossible. For* $a = 3$*, the same inequality would read*

$$25 \cdot 27 \leq 3 \cdot 3 \cdot 5^2,$$

which is false. For $a = 1$*, the condition is satisfied:*

$$25 + 8 \cdot 5 \cdot 2 = 105 \leq 147 = 3 \cdot 1 \cdot 7^2.$$

Hence

$$B_{\mathrm{int}}(8, 5, \sqrt{3}) = 2.$$

Moreover, the graph is not complete bipartite; for instance, $0 \in V_1$ *and* $4 \in V_2$ *are nonadjacent. Therefore a balanced bi-independent pair of size* 2 *exists. Consequently,*

$$\alpha_{\mathrm{bal}}(G) = 2.$$

So the integer-refined bound gives the exact value 2*, whereas the standard bound gives only* $\alpha_{\mathrm{bal}}(G) \leq 4$ *after parity rounding.*

5 The hypercube case

Let Q_d be the d-dimensional hypercube. We identify its vertices with subsets of $[d] = \{1, \dots, d\}$, and join two subsets when their symmetric difference has size 1. Let

$$X_0 = \{S \subseteq [d] : |S| \text{ is even}\}, \qquad X_1 = \{S \subseteq [d] : |S| \text{ is odd}\}.$$

Then Q_d is bipartite with bipartition $X_0 \sqcup X_1$.

A balanced independent set in Barber's sense is an independent set $I \subseteq V(Q_d)$ such that

$$|I \cap X_0| = |I \cap X_1|.$$

Since Q_d has no edges inside X_0 or inside X_1, such a set is exactly a balanced bi-independent pair (A, B) with $A \subseteq X_0$ and $B \subseteq X_1$. Therefore Barber's balanced-independent-set parameter is exactly $\alpha_{\text{bal}}(Q_d)$.

5.1 The cube isoperimetric input

For $A \subseteq X_0$, let $N(A) \subseteq X_1$ denote the set of odd vertices adjacent to at least one vertex of A. Order the vertices of Q_d by the simplicial order: $S < T$ if either $|S| < |T|$, or $|S| = |T|$ and S precedes T in lexicographic order. The following theorem is due independently to Bezrukov and to Körner–Wei; see also Tiersma.

Theorem 2 (Even-set isoperimetry in the cube)**.** *Let $A \subseteq X_0$, and let $I \subseteq X_0$ be the initial segment of the simplicial order restricted to X_0 with $|I| = |A|$. Then*

$$|N(I)| \leq |N(A)|.$$

Moreover, $X_1 \setminus N(I)$ is a terminal segment of the simplicial order restricted to X_1.

5.2 Barber's theorem

Theorem 3 (Barber)**.** *For every $d \geq 1$,*

$$\alpha_{\text{bal}}(Q_d) = \begin{cases} 2^{d-1} - 2\binom{d-2}{(d-2)/2}, & d \text{ even}, \\ 2^{d-1} - \binom{d-1}{(d-1)/2}, & d \text{ odd}. \end{cases}$$

Proof. The cases $d = 1, 2$ are immediate, so assume $d \geq 3$.

For a fixed $A \subseteq X_0$, the largest possible odd side of an independent set with even side A is $X_1 \setminus N(A)$. Thus a balanced independent set with $|A| = |B| = t$ exists if and only if there is some $A \subseteq X_0$ with $|A| = t$ and

$$t \leq |X_1 \setminus N(A)|.$$

By Theorem 2, for each fixed t it is enough to check the initial segment $I_t \subseteq X_0$ of size t: replacing A by I_t can only decrease the neighborhood and hence can only increase the available terminal odd set $X_1 \setminus N(I_t)$.

As t increases, the initial segment I_t increases and the terminal set $X_1 \setminus N(I_t)$ decreases. Hence if, for some initial segment I_t, we have

$$|I_t| = |X_1 \setminus N(I_t)|,$$

then that value of t is the largest feasible half-size.

It remains to identify the crossing point. We use the notation

$$[u,v]^{(q)} = \{S \subseteq \{u, u+1, \dots, v\} : |S| = q\},$$

and, when $D \subseteq [d]$ is disjoint from every set in a family $\mathcal{F}$,

$$D + \mathcal{F} = \{D \cup F : F \in \mathcal{F}\}.$$

Empty unions are ignored. A direct layer-by-layer check gives the following initial even segment A and the corresponding terminal odd set $B = X_1 \setminus N(A)$.

If $d = 4k$, take

$$\begin{aligned} A &= [d]^{(0)} \cup [d]^{(2)} \cup \cdots \cup [d]^{(2k-2)} \cup \big(\{1,2\} + [3,d]^{(2k-2)}\big), \\ B &= \big(\{1\} + [3,d]^{(2k)}\big) \cup [2,d]^{(2k+1)} \cup [d]^{(2k+3)} \cup \cdots \cup [d]^{(d-1)}. \end{aligned}$$

If $d = 4k+1$, take

$$\begin{aligned} A &= [d]^{(0)} \cup [d]^{(2)} \cup \cdots \cup [d]^{(2k-2)} \cup \big(\{1\} + [2,d]^{(2k-1)}\big), \\ B &= [2,d]^{(2k+1)} \cup [d]^{(2k+3)} \cup \cdots \cup [d]^{(d)}. \end{aligned}$$

If $d = 4k+2$, take

$$\begin{aligned} A &= [d]^{(0)} \cup [d]^{(2)} \cup \cdots \cup [d]^{(2k-2)} \cup \big(\{1\} + [2,d]^{(2k-1)}\big) \cup \big(\{2\} + [3,d]^{(2k-1)}\big), \\ B &= [3,d]^{(2k+1)} \cup [d]^{(2k+3)} \cup \cdots \cup [d]^{(d-1)}. \end{aligned}$$

Finally, if $d = 4k+3$, take

$$\begin{aligned} A &= [d]^{(0)} \cup [d]^{(2)} \cup \cdots \cup [d]^{(2k)}, \\ B &= [d]^{(2k+3)} \cup [d]^{(2k+5)} \cup \cdots \cup [d]^{(d)}. \end{aligned}$$

In each case A is an initial segment of X_0, B is the associated terminal set $X_1 \setminus N(A)$, and $|A| = |B|$. These equalities are checked from Pascal's identity, symmetry of binomial coefficients, and $\sum_{q=0}^{m} \binom{m}{q} = 2^m$.

The common value $|A| = |B|$ is therefore the largest feasible half-size. The corresponding total size $2|A|$ is

$$2^{d-1} - 2\binom{d-2}{(d-2)/2} \quad \text{if } d \text{ is even,}$$

and

$$2^{d-1} - \binom{d-1}{(d-1)/2} \quad \text{if } d \text{ is odd.}$$

This proves the theorem. $\square$

5.3 The Laurent–Polak–Vargas hypercube conjecture

Laurent, Polak and Vargas define a sequence $(a(m))_{m\ge 0}$ by

$$a(0) = 0, \qquad a(2s) = 2^{2s} - \binom{2s}{s}, \qquad a(2s+1) = 2a(2s).$$

With the convention above, $a(1) = 0$.

Theorem 4 (Hypercube formula)**.** *For every $d \geq 1$,*

$$\boxed{\alpha_{\mathrm{bal}}(Q_d) = a(d-1).}$$

Proof. Apply Theorem 3 and compare its formula with the definition of $a(m)$.

If $d = 2s+1$ is odd, then

$$\alpha_{\mathrm{bal}}(Q_d) = 2^{2s} - \binom{2s}{s} = a(2s) = a(d-1).$$

If $d = 2s$ is even, then

$$\begin{aligned}\alpha_{\mathrm{bal}}(Q_d) &= 2^{2s-1} - 2\binom{2s-2}{s-1} \\ &= 2\left(2^{2s-2} - \binom{2s-2}{s-1}\right) \\ &= 2a(2s-2) = a(2s-1) = a(d-1).\end{aligned}$$

This includes $d = 2$, since $a(1) = 2a(0) = 0$. The case $d = 1$ gives $\alpha_{\mathrm{bal}}(Q_1) = 0 = a(0)$. □

Thus the hypercube conjecture is an immediate consequence of the earlier cube result of Barber. In particular, the exact value of $\alpha_{\mathrm{bal}}(Q_d)$ is governed by the isoperimetry of the Boolean cube rather than by the integer-refined spectral bound of Theorem 1.

6 Final remarks

The spectral refinement in Theorem 1 strengthens the standard proof at exactly one point. The standard argument lower-bounds $\|M\mathbf{1}_B\|_2^2$ by the continuous estimate

$$\|M\mathbf{1}_B\|_2^2 \geq \frac{(ra)^2}{n-a}.$$

The proof above instead uses the exact minimum of the same sum of squares under the additional fact that the entries of $M\mathbf{1}_B$ are integer neighbor counts. This changes the necessary condition from

$$r^2a^3 \leq \lambda^2 a(n-a)^2$$

to the stronger condition

$$r^2a^3 + n\, j_a(n-a-j_a) \leq \lambda^2 a(n-a)^2.$$

The hypercube theorem, however, requires a different method: the extremal balanced independent sets in Q_d are determined by initial and terminal segments in the simplicial order on the Boolean lattice.

References

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